FBCRS Document Analysis & Recommendation Engine

User Manual — Version 8.5

Contents

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1. What this tool does

The engine scans a folder of documents and produces an Excel inventory that suggests, for each file, which records series it belongs to and what should happen to it.

For every file it reports:

- a **records series code** from your classification rulebook, or `UNCLASSIFIED`
- the **retention period** and calculated **expiry date**
- a **recommended action** — Destroy, Do Nothing, Manual Review, or Transfer

to Archives
- a **confidence level** and a plain-English explanation of why
- whether the file is an exact **duplicate** of another file in the scan
- up to three **alternate codes** when the top match is uncertain

It reads inside `.docx`, `.pdf`, `.xlsx`, `.xlsm`, `.pptx`, and `.txt` files. Other file types are still inventoried, but classified on filename alone.

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2. What this tool does *not* do

Read this section before acting on any output.

It does not delete, move, or modify a single file. The tool is read-only. Every action in the report is a *recommendation* for a human to approve. Nothing happens to your documents.

`DESTROY` is a suggestion, never an instruction. Do not run a bulk deletion from this report. Every destroy recommendation needs review by someone with the authority to approve disposition.

It does not detect near-duplicates. Only byte-for-byte identical files. Two versions of the same document with a one-word change are treated as unrelated.

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3. Installation

3.1 What you need

Two files, **in the same folder**:

```
C:\FBCRS\
├── FBCRS_Analyzer.exe          (the application)
├── FBCRS_Master_Full.xlsx      (the classification rulebook)
└── repair_rulebook.exe         (rulebook checker - optional)
```

The rulebook **must** be named exactly `FBCRS_Master_Full.xlsx` and **must** sit beside the program. This is the single most common setup mistake.

3.2 Where to install it

Install to a local drive — `C:\FBCRS\` — not a network share.

The program writes temporary checkpoint files thousands of times during a scan.

Running it from a mapped drive or UNC path (`Z:\...` or `\\SERVER\...`) causes intermittent "Access is denied" failures partway through a scan, and is dramatically slower. You can *scan* network folders; just don't *run* from one.

3.3 Python setup (skip if using the .exe)

Requires Python 3.10 or later. Install the dependencies once:

```
pip install customtkinter openpyxl pypdf python-docx python-pptx
```

Package	Needed for
`customtkinter`	the interface — **required**
`openpyxl`	reading the rulebook and writing reports — **required**
`pypdf`	reading inside PDF files — optional
`python-docx`	reading inside Word files — optional
`python-pptx`	reading inside PowerPoint files — optional

The three optional packages degrade gracefully. Without `pypdf`, for example, PDFs are still inventoried and classified on filename, just not on content.

3.4 Where reports are saved

Automatically, to the current user's Documents folder:

```
C:\Users\<YourName>\Documents\FBCRS_Reports\
```

This is resolved at runtime, so on a colleague's machine it saves to *their* Documents folder. If Documents isn't writable, the program falls back to `%LOCALAPPDATA%\FBCRS_Reports`, then to the system temp folder. The active path is displayed in small grey text at the bottom of the window.

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4. Quick start

1. **Close the rulebook.** `FBCRS_Master_Full.xlsx` must not be open in Excel. Excel locks the file and the program will refuse to start.
2. **Launch.** Double-click `FBCRS_Analyzer.exe`. If you are running from source instead, use:

```
python C:\FBCRS\FBCRS_Record_Inventory_V8_5.py
```

3. **Click "Browse Folder"** and choose the folder to scan. Subfolders are included automatically.
4. **Click "Start Analysis."** The status line reports progress as `Scanning files: 1,240 / 8,900`.
5. **Wait.** The report opens in Excel automatically when finished, and a summary dashboard shows elapsed time, files found, and files classified.

First run advice

Start with 30–50 files you already know the answers for. This tells you whether the rulebook is tuned correctly for your content before you commit to a multi-hour scan. Check the **Confidence Reason** column on those known files — if it says the match came from generic terms, the rulebook needs work, not the folder.

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5. Reading the report — all 21 columns

The report has one sheet, `Inventory`, with the top row frozen and filters enabled. A hidden sheet named `Lists` holds the dropdown values — leave it alone unless you're deliberately editing the vocabularies.

Identification

Col	Header	Contents
A	File Name	Filename with extension
B	Extension	Lowercase extension, e.g. `.docx`
C	Content Category	`Document`, `Spreadsheet`, `PDF`, or `Other`
D	Date Created	From the filesystem
E	Date Modified	From the filesystem — **drives the expiry calculation**
O	Full File Path	Complete path, clickable hyperlink

> **Note on Date Created:** Windows resets creation dates when files are copied or moved. A 2011 record copied to a new share in 2024 will show as created in 2024. Treat this column as unreliable for anything legal.

Classification

Col	Header	Contents
F	Code	Records series code, `UNCLASSIFIED`, or `N/A` for skipped files
G	Series Title	Series name from the rulebook
Q	Confidence	`Very High` / `High` / `Medium` / `Low` — **editable dropdown**
R	Confidence Reason	Plain-English explanation of the confidence level
S–U	Alternate Code 1–3	Runner-up codes, when genuinely competitive

Retention and disposition

Col	Header	Contents
H	Retention Period	Raw retention text from the rulebook, e.g. `CCY + 2 years`
I	Retention Expiry Date	Calculated date, or blank if it couldn't be determined
J	Disposition	Raw disposition text from the rulebook
K	Recommended Action	**Editable dropdown** — the decision to act on
L	Reason	**Editable dropdown** — why that action was recommended

Review flags

Col	Header	Contents
M	Duplicate Detected	`YES` if an identical file appeared earlier in the scan
N	SME Review Required	`YES` for low-confidence or unclassified files
P	Comments	Error details, skip reasons, and other notes

The three dropdown columns

Columns **K**, **L**, and **Q** are constrained dropdowns. Typing anything not on the list is rejected. Validation extends 500 rows past your data, so pasting in more rows keeps the dropdowns working.

Column K — Recommended Action

Value	Meaning
`DESTROY`	Retention has lapsed, or the file is an exact duplicate or empty
`DO NOTHING`	Still within its retention period — leave it alone
`MANUAL REVIEW REQUIRED`	The engine could not decide; a person must
`TRANSFER TO ARCHIVES`	The rulebook disposition calls for archival transfer

Column L — Reason

Value	Meaning
`WITHIN RETENTION`	Expiry date is in the future
`PASSED RETENTION`	Expiry date has passed
`DUPLICATE`	Byte-identical to a file seen earlier in this scan
`EMPTY/DRAFT`	Zero bytes, or a temporary Office lock file (`~$…`)
`UNSUPPORTED FORMAT`	File type that cannot contain records (image, video, archive)
`MISSING DISPOSITION RULE`	Retention lapsed but the rulebook has no disposition for that code
`NEEDS REVIEW`	Not confidently classified, or unreadable

Column Q — Confidence

`Very High`, `High`, `Medium`, `Low`. Editable so a reviewer can record their own assessment after inspecting a file.

Order of precedence

When several rules could apply, the first match wins:

1. **Duplicate** → `DESTROY` / `DUPLICATE`
2. **Unclassified** → `MANUAL REVIEW REQUIRED` / `NEEDS REVIEW`
3. **Expired, disposition known** → the rulebook's disposition / `PASSED RETENTION`
4. **Expired, no disposition** → `MANUAL REVIEW REQUIRED` / `MISSING DISPOSITION RULE`
5. **Low confidence** → `MANUAL REVIEW REQUIRED` / `NEEDS REVIEW`
6. **Otherwise** → `DO NOTHING` / `WITHIN RETENTION`

> **Important:** duplicate detection sits at the top. A file that is both unclassified *and* a duplicate is marked `DESTROY`, not sent for review. If you want unclassified duplicates reviewed instead, filter on `Reason = DUPLICATE` **and** `Code = UNCLASSIFIED` before acting.

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6. How classification works

6.1 The evidentiary model

Each file's name and content are evaluated against a **controlled taxonomy** maintained in the rulebook: single-word **index terms** and compound **descriptors** associated with each records series. Every match contributes weighted evidence toward that series:

Evidentiary signal	Weight
Index term match	2
Compound descriptor match	10
Match located in the **file name**	× 3

A compound descriptor carries five times the weight of a lone index term because specificity scales with it: "financial statements" identifies a series far more reliably than "financial" or "statements" in isolation. A match located in the file name carries triple weight because a file name is a deliberate label a person assigned, whereas body content is frequently generic.

Inverse-frequency weighting. Each term's contribution is divided by the number of series that claim it — the same principle underlying inverse-document-frequency weighting in information retrieval. A term unique to a single series retains its full weight; one claimed by twenty series contributes only a twentieth of it. This correction is what makes the model reliable at scale: the rulebook currently holds **249 records series described by 3,272 index terms and descriptors combined**. At that volume, generic language inevitably recurs across dozens of unrelated series — without inverse-frequency correction, that overlap would dominate the arithmetic and erase any genuine signal. The correction is precisely what lets a taxonomy this large remain discriminating rather than collapsing toward its most common terms.

Terms claimed by more than twenty series are excluded from scoring entirely, on the basis that they carry no discriminative value at this scale.

Each distinct term is credited once per field, irrespective of repetition. A document restating "financial" four hundred times earns no more evidentiary weight than one stating it once — frequency within a document is not mistaken for relevance.

6.2 Confidence bands

Aggregate score	Confidence	Typically indicates
30 or more	Very High	An exact descriptor matched in the file name
12–29	High	Several distinctive terms matched
4–11	Medium	Moderate evidentiary support
below 4	Low	Only weak or non-discriminative terms matched

Confidence is downgraded to `Medium` when two series score within 15% of one another — a near-tie does not constitute high confidence, regardless of the raw score.

6.3 Abstention criteria — when the engine declines to classify

A file is returned as `UNCLASSIFIED` when the accumulated evidence does not meet a minimum evidentiary threshold:

- **no evidence found** → "*No rulebook keywords or phrases found — context is not clear*"
- **score below 2** → "*Only generic terms matched, shared across many series*"
- **score below 6 with no file-name support** → "*Weak match in document body only, nothing in the file name*"

Expect a meaningful proportion of `UNCLASSIFIED` results, and treat that as correct operation rather than a shortfall. Declining to classify on insufficient evidence is a deliberate design choice. A model that assigns a series to every file is not thereby more accurate — it simply conceals its own uncertainty, which is considerably more hazardous when the output informs destruction decisions.

6.4 Alternate code candidates (columns S–U)

A runner-up series is surfaced only when it is genuinely competitive with the leading result: within 40% of the top score, and reaching a minimum evidentiary threshold of 2 in its own right.

Consequently, **high-confidence determinations typically carry no alternate candidates**, which is by design. Surfacing the second, third, and fourth-place series unconditionally would present reviewers with three plausible-looking incorrect options alongside every clear-cut match — and in practice, reviewers select them.

Alternate candidates are always blank for `UNCLASSIFIED` files: if the leading result reflects noise rather than evidence, so do the candidates beneath it.

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7. Retention and expiry dates

7.1 Supported formats

The engine reads two patterns from the rulebook's Retention column:

Pattern	Meaning	Expiry
`CCY + N`	Current Calendar Year plus N years	31 December, year + N
`CFY + N`	Current Fiscal Year plus N years	31 March, fiscal year end + N

Fiscal year is treated as April–March: files modified January to March fall in the fiscal year ending that same calendar year; April onwards fall in the fiscal year ending the next.

Surrounding text is ignored, so `CCY + 2 years`, `CCY+2`, and `ccy + 2` all parse identically.

7.2 Important limitations

Expiry is calculated from Date Modified. Records retention normally runs from file *closure* — case closed, project completed, employee departed — which the filesystem does not know. A file edited last week shows a retention clock that started last week, regardless of the underlying record's actual age.

Retention text the engine can't parse produces no expiry date. Entries like `Within a maximum of 1 month` or `Superseded + 1` are left blank, and the file is reported as `WITHIN RETENTION` — indefinitely. Blank expiry dates are not a guarantee of currency; they mean *unknown*. Filter on blank column I to find these.

Decimals are truncated. `CCY+0.3` parses as `CCY+0`, giving an expiry of 31 December in the year the file was modified. Review any rulebook entries with fractional retention.

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8. Duplicate detection

8.1 How it works

Duplicates are found in two stages. First, file **sizes** are compared — identical files must have identical sizes, so this eliminates almost everything without reading any file content. Only files whose size collides with another are then **hashed** (SHA-256 over the first 5 MB).

8.2 Which copy gets flagged

Files are processed in sorted path order, and **the first copy encountered is kept**; later copies are marked `DUPLICATE` / `DESTROY`. This is deterministic — re-scanning the same folder flags the same copies every time.

> **"First in sorted order" is alphabetical, not a judgement about which copy is the real record.** `Archive\report.docx` sorts before `Current\report.docx`, so the archived copy would be kept and the current one marked for destruction. Always confirm which copy you actually want before acting on a duplicate.

8.3 What is not detected

- **Near-duplicates.** One changed word makes two files unrelated to this tool.
- **Files differing beyond the first 5 MB.** Two large files with identical first 5 MB and identical total size are reported as duplicates. Rare, but possible with padded or templated files.
- **Duplicates among skipped types.** Images, videos, and archives are never hashed, so identical copies of them are not flagged.
- **Duplicates across separate scans.** Detection is per-scan only.

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9. The review workflow

A practical order for working through a report.

Step 1 — Fix the plumbing first. Filter column P (Comments) for `ACCESS DENIED` and `Scan failed`. These files were never actually examined. Resolve permissions and re-scan before drawing conclusions from the totals.

Step 2 — Clear the safe wins. Filter column L for `EMPTY/DRAFT`. Zero-byte files and `~$` Office lock files are genuinely disposable and need little scrutiny.

Step 3 — Review every `DESTROY` recommendation. Filter column K for `DESTROY`. Sort by column M so duplicates group together. For each, confirm the copy being kept is the one you want. This is where irreversible mistakes happen.

Step 4 — Work the review queue by confidence. Filter column N for `YES`, then sort column Q ascending so `Low` appears first. Use columns R, S, T, and U — Confidence Reason tells you *why* it's uncertain, and the alternates give you candidate codes to pick from instead of searching the full series list.

Step 5 — Find the silent gaps. Filter for blank column I (no expiry date) and for `MISSING DISPOSITION RULE`. These are rulebook problems masquerading as file problems, and they will recur on every future scan until fixed in the rulebook.

Step 6 — Record your decisions in the dropdowns. Override columns K, L, and Q as you review. The constrained vocabularies keep the report analysable afterwards — you can pivot it, count it, and hand it on.

> **Keep the original report unedited.** Save your reviewed version under a new name. Reports are timestamped, so a re-scan won't overwrite anything, but an unedited baseline is worth having.

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10. Maintaining the rulebook

10.1 Structure

`FBCRS_Master_Full.xlsx` needs four sheets, each with a header row:

Sheet	Column A	Column B	Columns C–D
Records	Code	Series Title	Retention, Disposition
Keywords	Code	Keyword (one per row)	—
Phrases	Code	Phrase (one per row)	—
Synonyms	Term	Replacement	—

Additional sheets are ignored, so a `FunctionCodes` reference sheet or similar can live in the same workbook safely.

10.2 Editing

1. Close the app.
2. Edit the workbook and save it.
3. **Save and close the workbook.**
4. Restart the app — the rulebook is read once at startup.

10.3 Writing effective keywords

Specific beats general. `reconciliation` identifies a series; `report` identifies nothing. Rarity weighting means generic terms contribute almost nothing anyway, so they're wasted effort.

Phrases outperform keywords — 10 points versus 2, and far fewer false positives. `key performance indicator` is worth more than `key` plus `performance` plus `indicator` separately.

Terms are matched as whole words. `plan` will not match `plans` or `planning`. Add each form you need. Conversely this is protective: `bi` matches only the standalone word, never the inside of `liability`.

Never put multi-line content in a Synonyms cell. Each synonym replacement must be a short single-line phrase. A pasted block in that column expands into every document the engine reads and destroys all classification accuracy — see section 12.

10.4 Auditing the rulebook

`repair_rulebook.exe` checks a rulebook and reports problems.

Double-click it. A file dialog opens — select the rulebook you want checked. The findings appear in a console window, which stays open until you press Enter.

You can also pass the path directly if you prefer:

```
repair_rulebook.exe "C:\FBCRS\FBCRS_Master_Full.xlsx"
```

What it reports

- **Corrupt Synonyms entries** — multi-line or overlong replacement values, the fault that makes every file classify as the same code
- **Over-shared keywords** — terms claimed by so many series that they carry no signal, listed worst first
- **Over-shared phrases** — the same problem in the Phrases sheet
- **How many keywords are genuinely distinctive** — owned by exactly one series. This is the single best measure of rulebook health

What it writes

Two files, beside the rulebook you selected:

File	Contents
`<name>_REPAIRED.xlsx`	The cleaned rulebook — **use this one**
`<name>_AUDIT.txt`	The full report, saved so you can keep or circulate it

The original is never modified. Records, Keywords, and Phrases are copied through untouched; only the Synonyms sheet is repaired, and anything removed from it is moved to a `FunctionCodes` sheet rather than deleted, so nothing is lost.

To put a repaired rulebook into service, rename it to `FBCRS_Master_Full.xlsx` and place it beside `FBCRS_Analyzer.exe`.

Run this whenever you make significant rulebook edits, and any time classification quality drops.

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11. Performance tuning

Expect roughly **30–40 ms per file** on local storage. Network shares are slower and vary with the link.

If scans are too slow, adjust these constants near the top of the script, in this order of impact:

```
XLSX_ROWS = 500     # rows read per spreadsheet sheet
TEXT_CAP  = 100000  # characters of content fed to the matcher
PDF_PAGES = 10      # pages read per PDF
```

`XLSX_ROWS` is the biggest lever. Parsing spreadsheets dominates runtime. Dropping it to `150` roughly halves total scan time and rarely changes results — if a spreadsheet's subject isn't evident in its first 150 rows and its filename, another 350 rows seldom help.

Other settings:

Constant	Default	Effect
`MAX_WORKERS`	`min(8, cores × 2)`	Parallel threads. Raising helps on high-latency shares
`MAX_SIZE_BYTES`	15 MB	Files above this are classified on filename only
`HASH_BYTES`	5 MB	Bytes hashed for duplicate comparison
`DOCX_PARAS`	200	Paragraphs read per Word file
`PPTX_SLIDES`	10	Slides read per presentation

General advice

- **Run from a local drive.** The single biggest factor.
- **Break very large directories into sub-folders** and scan separately. Reports are named after the folder, so they won't collide.
- **Scan outside business hours** for large network folders.

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12. Troubleshooting

"Fatal Error: Missing Rulebook"

`FBCRS_Master_Full.xlsx` is not in the same folder as the program. The dialog shows the exact folder being searched. Check for a renamed file (`FBCRS_Master_Full_REPAIRED.xlsx` won't be found) or a hidden double extension (`.xlsx.xlsx`).

"Fatal Error: Rulebook Read Error"

Usually the rulebook is open in Excel. Close it. If it persists, the file may be corrupt — open it in Excel and re-save.

"Access is denied" partway through a scan

You're running the program from a network share. Move it to `C:\FBCRS\`. See section 3.2.

"The output Excel file is open"

A previous report is open in Excel. Close it. The program retries three times before giving up.

Every file gets the same code

A corrupt **Synonyms** sheet. If a synonym's replacement value contains multiple lines or a long block of text, that block gets injected into every document the engine reads, so every file matches the same series.

Run `repair_rulebook.exe` (section 10.4) and use the repaired copy. Verify by checking the **Confidence Reason** column — if unrelated files all claim to match the same phrase, this is the cause.

Nearly everything is UNCLASSIFIED

Check **Confidence Reason** to distinguish two very different situations:

- "*No rulebook keywords or phrases found*" → your rulebook lacks terms for this content. Add keywords for the series that actually appear in this folder.
- "*Only generic terms matched*" → the rulebook's terms are too generic. Add distinctive phrases.

If review volume is genuinely unmanageable, `MIN_SCORE` and `BODY_ONLY_MIN` control the thresholds — but read the evidence before loosening them. Lowering thresholds doesn't create accuracy, it just converts visible uncertainty into invisible uncertainty.

Files are missing from the report

Files and folders whose names begin with `~` or `.` are skipped by design (Office temp files, hidden folders). Also check column P for `ACCESS DENIED`.

Where to look when nothing else explains it

```
Documents\FBCRS_Reports\FBCRS_error_log.txt
```

Full technical tracebacks for any failure, with timestamps. Also check for leftover `*_temp.csv` files in that folder — if a scan was interrupted, the partial results survive there and can be opened in Excel.

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13. Building and sharing an .exe

13.1 Building

```
pip install pyinstaller
cd /d C:\FBCRS
```

The main application — `--windowed` suppresses the console, since it has its own interface:

```
pyinstaller --onefile --windowed --name FBCRS_Analyzer
FBCRS_Record_Inventory_V8_5.py
```

The rulebook checker — `--console` here, *not* `--windowed`, because its output is the console window. Build it windowed and the findings go nowhere:

```
pyinstaller --onefile --console --name repair_rulebook repair_rulebook.py
```

Both results appear in `C:\FBCRS\dist\`.

If `FBCRS_Analyzer.exe` builds but crashes on launch with a `customtkinter` path error, rebuild it with `--collect-all customtkinter` added.

13.2 Packaging for colleagues

The rulebook is deliberately **not** bundled into the exe, so you can update keywords without rebuilding. Ship a folder:

```
FBCRS_Analyzer\
├── FBCRS_Analyzer.exe          (the application)
├── FBCRS_Master_Full.xlsx      (the rulebook)
├── repair_rulebook.exe         (rulebook checker)
└── FBCRS_User_Manual.pdf       (this manual)
```

Zip it. Tell recipients to unzip it and keep the files together — the exe cannot run without the rulebook beside it.

13.3 What to expect on their machines

Reports save to their own Documents folder, not yours. Nothing is hardcoded to your user profile.

Windows SmartScreen and antivirus will likely complain. Unsigned PyInstaller executables routinely trigger "Windows protected your PC" warnings and are sometimes quarantined outright. This is a known false positive, but recipients will need to click through it — and on a locked-down corporate machine it may be blocked entirely. Warn people in advance, and check with IT before wide distribution.

Test on a machine without Python installed before sending it widely. An exe that works on the build machine can still be missing components that only surface on a clean one.

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14. Known limitations

Classification is keyword matching, not comprehension. No understanding of context, business purpose, or the difference between an official record and a working copy.

Retention runs from Date Modified, not file closure. See section 7.2.

Unparseable retention text produces no expiry date and a `WITHIN RETENTION` verdict. Blank expiry means *unknown*, not *current*.

Duplicate "keeper" selection is alphabetical. Not a judgement about which copy is authoritative. See section 8.2.

Only the beginning of each file is read — 10 PDF pages, 200 Word paragraphs, 500 spreadsheet rows per sheet, 100,000 characters total. Content past those limits is invisible. Configurable, at a cost in speed.

Scanned documents and images are not OCR'd. A scanned PDF with no text layer yields no content, and classifies on filename alone.

Files over 15 MB are classified on filename only.

Encrypted and password-protected files cannot be read. They appear with `Scan failed` in Comments.

Very long paths may fail on Windows. Paths beyond 260 characters can produce `ACCESS DENIED` unless long path support is enabled. Excel also refuses hyperlinks over 255 characters, so those cells are left as plain text.

Duplicate detection is per-scan. No memory between runs.

Date Created is unreliable after files are copied or moved.

Appendix A — configuration reference

All constants sit near the top of the script.

Scoring

Constant	Default	Purpose
`KW_BASE`	2.0	Points per keyword match
`PH_BASE`	10.0	Points per phrase match
`FN_BOOST`	3.0	Multiplier for filename matches
`MAX_OWNERS`	20	Terms claimed by more series than this are ignored
`MIN_SCORE`	2.0	Below this → `UNCLASSIFIED`
`BODY_ONLY_MIN`	6.0	Threshold when nothing matched in the filename
`CONF_BANDS`	30 / 12 / 4	Very High / High / Medium cutoffs

Alternates

Constant	Default	Purpose
`N_ALTERNATES`	3	Number of runner-up columns
`ALT_MIN_SCORE`	2.0	Minimum score for a runner-up to be shown
`ALT_REL_MIN`	0.40	Runner-up must reach this fraction of the winner. Set to `0.0` to always show three

Reading and performance

Constant	Default	Purpose
`MAX_WORKERS`	`min(8, cores × 2)`	Parallel worker threads
`TEXT_CAP`	100000	Max characters fed to the matcher
`XLSX_ROWS`	500	Rows read per sheet — **largest speed lever**
`XLSX_SHEETS`	3	Sheets read per workbook
`PDF_PAGES`	10	Pages read per PDF
`DOCX_PARAS`	200	Paragraphs read per Word file
`PPTX_SLIDES`	10	Slides read per presentation
`HASH_BYTES`	5 MB	Bytes hashed for duplicate comparison
`MAX_SIZE_BYTES`	15 MB	Above this, filename-only classification

File handling

Constant	Contents
`SKIP_EXTS`	`.jpg .jpeg .png .gif .bmp .tiff .mp4 .mov .avi .wmv .mkv .zip .7z .rar .exe .dll .msi`
`ACTION_OPTIONS`	The four Recommended Action values
`REASON_OPTIONS`	The seven Reason values
`CONF_OPTIONS`	The four Confidence values

Adding a value to `ACTION_OPTIONS`, `REASON_OPTIONS`, or `CONF_OPTIONS` automatically extends the corresponding Excel dropdown.

> **Note:** `EXTS` exists in the code but is not used for filtering. Every file not in `SKIP_EXTS` is inventoried; content extraction simply only works for the supported types.

Appendix B — file inventory

File	Purpose
`FBCRS_Analyzer.exe`	The application
`FBCRS_Master_Full.xlsx`	Classification rulebook — must sit beside the app
`repair_rulebook.exe`	Rulebook checker and Synonyms repair tool

Source files, if you are running or rebuilding from Python rather than the exes:

File	Builds
`FBCRS_Record_Inventory_V8_5.py`	`FBCRS_Analyzer.exe`
`repair_rulebook.py`	`repair_rulebook.exe`

Generated output

Location	Contents
`Documents\FBCRS_Reports\<folder> summary <timestamp>.xlsx`	The report
`<rulebook>_REPAIRED.xlsx`	Cleaned rulebook from `repair_rulebook.exe`
`<rulebook>_AUDIT.txt`	Rulebook audit report
`Documents\FBCRS_Reports\<folder>_<timestamp>_temp.csv`	Checkpoint; deleted on success, survives a crash
`Documents\FBCRS_Reports\FBCRS_error_log.txt`	Technical error log, appended to

Version 8.5. This manual describes the tool's behaviour as built; it is not a records management policy document. Disposition decisions remain the responsibility of the reviewing records professional.